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Automation and Political Behavior

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Abstract

Automation and Political Behavior

We investigate how automation shapes two distinct dimensions of political behavior, voter turnout and conservative party vote share. We estimate the causal effect of robot exposure using a shift-share instrumental variable strategy that interacts 1995 local industry shares with robot adoption patterns in Singapore, which serve as an exogenous source of identification. We find that greater exposure to industrial robots significantly depresses voter turnout and shifts support toward conservative parties. These effects are concentrated in regions with higher immigrant share. Our findings provide the first causal evidence on the political impact of technological change in Korea, suggesting that automation reshapes political behavior through mechanisms similar to those documented in Western contexts.

Keywords : automation, technological change, voting behavior, political economy, labor markets

Chapter 1. Introduction

Technological advances have repeatedly reshaped economic structures throughout history, from power looms in the early 19th century to computerization in the late 20th century. These economic transformations generate aggregate productivity gains, yet they invariably create losers alongside winners (Autor et al., 2003; Goos & Manning, 2007; Autor & Dorn, 2013; Acemoglu & Restrepo, 2019, 2020). Since workers' skills constitute primary assets, the social costs of automation, including income loss, forced displacement, and skill obsolescence, threaten not only the livelihoods of displaced workers but also the stability of incumbent political leaders (Acemoglu & Robinson, 2013). Workers who bear the brunt of automation may therefore resort to non-market forms of resistance, including political activism (Mokyr, 1998; Mokyr et al., 2015).

This study investigates the relationship between automation and voting behavior in South Korea. Existing research has extensively documented the labor market consequences of automation (Graetz & Michaels, 2018; Acemoglu & Restrepo, 2020; Dauth et al., 2021; Kim, 2024). Yet despite the scale of these disruptions, research examining the political consequences of automation remains scant. The existing evidence, while suggestive, has largely centered on support

for nationalism and radical-right parties (Anelli et al., 2019), often over a short time horizon (Frey et al., 2018). In particular, the effects of automation on electoral participation and conservative vote share remain underexplored.

We construct regional robot exposure measures at the county-level (si-gun-gu level) using industrial robot adoption data from the International Federation of Robotics (IFR) combined with local employment structure. The main identification challenge stems from the potential endogeneity of regional robot exposure. Local industrial policies may influence technology adoption decisions, and unobserved regional characteristics may correlate with both automation exposure and voting patterns. We address this concern using a shift-share instrumental variable following Acemoglu and Restrepo (2020), constructed by interacting two components. The shift component captures robot adoption patterns in Singapore, reflecting global technological trends independent of Korean political dynamics, while the share component employs predetermined Korean regional employment structure. This interaction generates plausibly exogenous variation in local automation exposure, driven by differential regional exposure to global technology shocks across regions with varying industrial structures. We examine two key political outcomes, voter turnout and conservative two-party vote share, across presidential elections from 2007 to 2022.

We find that increased robot exposure significantly reduces voter turnout and increases conservative two-party vote share, a pattern broadly consistent with findings from other developed countries. These results persist when controlling

for demographic characteristics and including region and year fixed effects. Heterogeneity analysis further reveals that these effects are statistically significant only in regions with high immigrant share, while the effects are insignificant in regions with low immigrant share.

These findings carry implications beyond a single case. That systematic effects of robot exposure on both voter turnout and partisan preferences emerge even after addressing confounding structures through a shift-share instrumental variable design strengthens confidence in the causal interpretation of these relationships. The political institutions of the context studied differ substantially from those of the Western democracies where the political consequences of automation have been most extensively studied, yet a broadly similar pattern nonetheless emerges. Moreover, the concentration of these effects in regions with high immigrant share suggests that psychological and cultural responses to economic threat, of the kind documented in Western contexts, may operate as transmission channels as well, pointing to mechanisms that generalize across diverse institutional settings.

The remainder of this paper is structured as follows. Chapter 2 reviews the relevant literature, beginning with the labor market consequences of automation and the emerging literature on automation and political outcomes, before turning to the Korean context, including the regional cleavage structure that motivates the identification strategy employed in this paper. Chapter 3 describes our data sources and empirical strategy, including the construction of regional robot

exposure measures and our shift-share instrumental variable strategy. Chapter 4 presents the main results and robustness checks. Chapter 5 concludes with a discussion of implications for understanding technological change and political dynamics in developed economies.

Chapter 2. Literature Review

1. Automation and Labor Market

The impact of technological change on employment and wages has not been settled, and the current wave of automation is no exception. For some, the spread of computer numerical control machinery, industrial robots, and artificial intelligence represents something qualitatively new, a force capable of displacing human labor faster than economies can generate replacement work, and thus a harbinger of structural joblessness on a scale without historical precedent. For others, the appropriate lens is the longer arc of technological history, in which each wave of mechanization has ultimately expanded rather than contracted the demand for human labor, even as it reshuffled which kinds of work humans performed.

The historical record offers reasons for both views. In the Industrial Revolution, the mechanization of spinning and weaving displaced artisans, triggering the Luddite uprisings as workers destroyed machinery in a desperate attempt to halt a transformation already underway (Mokyr, 1992; Mantoux, 2013). The mechanization of agriculture, from horse-powered reapers in the nineteenth

century to tractors and combine harvesters in the twentieth, displaced large numbers of workers from land-based labor that had structured rural life for centuries (Rasmussen, 1982; Olmstead & Rhode, 2001). However, history also records the countervailing force. Each prior wave of displacement eventually gave rise to new occupations that absorbed the labor it had displaced, from the line workers, engineers, and managers of the factory system to the programmers, analysts, and technicians of the computer revolution (Chandler, 1977; Mokyr, 1992; Goldin & Katz, 2008).

Today too we are witnessing a period of rapid automation, and a growing body of empirical work has begun to map its consequences for labor markets across countries and analytical levels (Acemoglu & Restrepo, 2020; Acemoglu et al., 2020; Dauth et al., 2021; Koch et al., 2021). The evidence reveals that the effects of automation depend critically on whether the jobs and tasks displaced by machines are offset by new labor demand generated elsewhere. In the United States, robot exposure in local labor markets led to significant reductions in both the number of jobs and real wages (Acemoglu & Restrepo, 2020). Firm-level evidence from France tells a more nuanced story. Firms that purchase robots increase their own employment, value added, and productivity, but the aggregate effect at the industry level is negative because non-adopting firms contract, offsetting the gains (Acemoglu et al., 2020). Germany presents a different pattern. Robot exposure reduces manufacturing employment, but this loss is offset by the creation of new jobs in services, with managers and engineers as the primary

beneficiaries (Dauth et al., 2021). In Spain, better-performing firms are more likely to adopt robots, and adoption generates substantial gains for these firms, including a reduction in labor cost share and net job creation of roughly ten percent within four years (Koch et al., 2021). Across these cases, a common pattern emerges. The gains from automation tend to concentrate among high-skill workers, high-performing firms, and sectors capable of generating new tasks, while the costs fall disproportionately on workers in routine and manufacturing occupations whose displacement is not reliably offset by new employment opportunities elsewhere.

Beyond the aggregate effects documented above, the distributional consequences of automation are deeply unequal. Growing evidence suggests that automation has not only reduced employment and wages in aggregate but has concentrated its negative effects on workers in the middle of the wage and skill distribution, producing a pattern of employment and wage polarization (Goos et al., 2009; Acemoglu & Autor, 2011; Autor & Dorn, 2013; Acemoglu & Restrepo, 2022).

One explanation for why automation has targeted middle-skill workers in particular points to the nature of the tasks they perform. Autor (2014, 2015) argues that many manual and abstract tasks embed rich tacit knowledge, skills that workers possess but cannot fully articulate, making them difficult to automate. Routine tasks, by contrast, are technologically easier to codify and automate, and because these tasks are disproportionately performed by middle-skill workers,

automation has had its most adverse effects precisely on this group. Acemoglu and Loebbing (2026) offer a complementary explanation grounded in profitability. Automation has concentrated on middle-skill tasks not only because they are routine but because they are the most profitable ones to automate, as low-skill tasks can be performed at sufficiently low labor cost that the cost advantage of capital over humans is reduced. Under the conditions they identify, specifically when wages at the bottom of the distribution are already sufficiently low, further automation leads to both wage and employment polarization. These distributional consequences are further compounded by the timing of displacement. Jaimovich and Siu (2020) document that routine occupation employment declines are concentrated in recessions, and that the sluggish aggregate employment recoveries observed after downturns reflect the permanent disappearance of these routine jobs rather than a temporary cyclical contraction.

South Korea, one of the world's most robotized economies, offers a particularly instructive case of how these patterns play out under conditions of intensive and rapid robot adoption. Using regional variation in robot exposure constructed from national industry-level adoption data and the initial distribution of industrial employment across cities, Kim (2024) finds that robots did not reduce aggregate local labor demand, but that this stability conceals concentrated losses. Robot exposure is associated with declines in job vacancies in the manufacturing sector and routine jobs, with no significant relationship found in the service sector or non-routine jobs.

The absence of offsetting job creation in services or non-routine employment suggests that new employment opportunities have not materialized at the regional level in Korea. Using individual-level data, Adema et al. (2025) find that the early wave of robot adoption from 2005 to 2009 concentrated its displacement effects on low-educated workers, who experienced job switching and a rise in part-time employment. From 2009 onward, while vocationally and highly educated workers benefited from wage increases, low-educated workers were excluded from these gains, consistent with displacement effects dominating at the individual level. Taken together, Korean evidence reinforces a pattern observed across multiple national contexts. The costs of automation fall disproportionately on lower-skilled workers in manufacturing and routine occupations, while the gains from productivity accrue elsewhere, a distributional asymmetry that forms the economic backdrop against which this paper examines political behavior.

One possible explanation for this distributional asymmetry, and for the relative absence of offsetting employment gains in the Korean context, lies in the structural features of the Korean labor market. The labor market in Korea is characterized by a pronounced dual structure, in which regular workers at large firms enjoy strong union protection and relative employment security, while subcontract and non-regular workers in the same production networks occupy a structurally vulnerable position. When robots are introduced, the adjustment costs tend to fall on this latter group. Lower-educated, non-regular workers are displaced through job switching and transitions into part-time employment, while

the productivity gains accrue to incumbent regular workers and capital. Reinstatement into stable employment is correspondingly difficult for this population. Workers who benefit from new task creation are typically higher-educated workers with the skills and credentials to access emerging occupations. Displaced subcontract workers, by contrast, face structural barriers to reentry, as the new jobs generated by automation-driven productivity growth are concentrated in sectors and firm tiers to which workers at the periphery of the production system have limited access. This structural account remains speculative in the absence of direct evidence on the employment trajectories of subcontract versus regular workers under robot exposure, but it is consistent with the pattern documented above and suggests that the dual labor market structure of Korea may amplify the distributional consequences of automation in ways that are not fully captured by aggregate measures of local labor demand.

2. Korean Context

The democratic era of Korea began in 1987 when Roh Tae-Woo was elected to the presidency following mass mobilization and demands for direct presidential elections. The Roh administration was then followed by the presidencies of Kim Young-Sam and Kim Dae-Jung, who won elections in 1992 and 1997 respectively. The competitive politics that emerged from this transition have been structured

around a persistent regional cleavage, particularly between Jeolla in the southwest and Gyeongsang in the southeast. Conservative parties have drawn their core electoral support from Gyeongsang, while progressive parties have consistently dominated in Jeolla, with each region delivering overwhelming majorities to their respective partisan camp (Horiuchi & Lee, 2008).

Gyeongsang is also home to the manufacturing heartland of Korea, and it is precisely this manufacturing base that became the primary target of successive waves of industrial automation policy in Korea. Following the Asian financial crisis, the state introduced its first official support program for the industrial robot industry in 2005. The 2009 Development and Distribution of Intelligent Robots Act then expanded this support to actively encourage firms to adopt automation in their production processes. By 2019, the cumulative effect of these policies had positioned South Korea among the most heavily robotized economies in the world, with a robot stock per worker exceeding that of Germany and Japan by more than a factor of two (Adema et al., 2025). As Figure 1 illustrates, the initial share of manufacturing workers across Korean cities, measured in 2007, is far from uniform. Regions with a greater manufacturing base tend to be the same regions that later experienced higher robot exposure, and these areas are disproportionately concentrated in the southeast, including much of Gyeongsang, which has historically anchored conservative electoral support.

[Figure 1]

This overlap, in which regions with historically strong conservative electoral support coincide substantially with the areas most exposed to industrial automation, creates a fundamental identification challenge for studying the political consequences of automation. Any observed correlation between robot exposure and political outcomes reflects at least two distinct processes that are difficult to disentangle. The first is the pre-existing regional political alignment, and the second is any causal effect that automation itself may have had on political behavior. Distinguishing between these two processes is the central identification problem this paper addresses. The shift-share instrumental variable design employed below exploits variation in robot exposure driven by national-level adoption trends interacted with industrial composition, isolating the component of local robot exposure that is plausibly exogenous to the regional political legacies documented above.

3. Automation and Political Behavior

1) Political Engagement

The labor market consequences of automation reviewed in the preceding sections raise a question that lies outside the scope of labor economics alone: how do workers and communities exposed to automation-induced economic

disruption respond politically? At an intuitive level, such disruption might be expected to mobilize affected workers (Schlozman & Verba, 1979; Kern et al., 2015). Yet a growing body of empirical evidence suggests that automation-induced economic risk suppresses rather than activates political engagement, with affected workers more likely to withdraw from participation than to mobilize (Kurer & Häusermann, 2022; Gonzalez-Rostani, 2024; Bratsberg et al., 2025).

One reason why automation tends to suppress rather than activate political engagement lies in the nature of the shock itself. Unlike trade policies such as tariffs or free trade agreements, which have identifiable political authors, automation lacks a clear villain, making it difficult for workers to attribute responsibility for their economic losses to political elites (Gonzalez-Rostani, 2024). Technology spreads rapidly and appears inevitable, and it exhibits a dual nature, raising efficiency while simultaneously threatening jobs. This ambivalence makes it difficult to frame automation as unambiguously harmful, further undermining workers' capacity to assign political blame. Moreover, unlike acute economic shocks that are temporally bounded, automation-induced displacement is structural and cumulative in nature, unfolding gradually across industries and regions over extended periods rather than as a discrete, datable event (Autor et al., 2003; Acemoglu & Restrepo, 2019, 2020). This makes it particularly difficult for affected workers to connect their deteriorating economic circumstances to the political arena, let alone to hold specific actors accountable (Kurer et al., 2019).

Beyond the attributional challenge, automation suppresses political engagement through more direct channels. The economic losses it generates, including wage declines, job displacement, and reduced reemployment opportunities, erode the material resources that political participation requires. Workers whose earnings and time are severely constrained by displacement are less able to engage in costly political activities, whether voting, canvassing, or donating (Brady et al., 1995).

Even among workers who have not yet experienced outright displacement, the anticipation of job loss can erode political efficacy, the belief that political participation can produce meaningful change in one's circumstances. Workers preoccupied with economic insecurity become less attentive to politics and less confident in their capacity to influence outcomes, producing disengagement rather than mobilization (Gonzalez-Rostani, 2024). Because the structural characteristic of automation means that the anticipation of job loss extends well beyond those who have already lost employment, its disengaging effects may be correspondingly broad.

2) Political Preferences

The political preferences of workers exposed to automation present a paradox that standard political economy struggles to resolve. Under standard models of rational political behavior, workers exposed to economic risk should demand

greater redistribution and support parties that offer compensatory policies, and indeed several studies confirm that automation exposure raises preferences for government intervention and redistribution (Margalit, 2013; Thewissen & Rueda, 2019). Yet the electoral evidence points in a different direction. A growing body of research shows that exposure to automation is associated with greater support for nationalist, populist, and conservative parties rather than parties advocating redistribution (Frey et al., 2018; Anelli et al., 2019; Im et al., 2019).

One possible explanation is that the electoral distinction between redistributive and non-redistributive parties has become increasingly blurred. As mainstream left and right parties have converged on key redistribution issues (Kriesi, 1998, 2012; Berman & Kundnani, 2021), voters exposed to automation may find it difficult to identify a credible partisan alternative capable of addressing their economic concerns. Although these arguments were developed primarily in Western countries, similar trends can also be observed in Korea. Existing studies suggest that Korean voters often struggle to distinguish between major parties on economic and redistributive issues, weakening the link between socioeconomic interests and partisan choice (Moon, 2017). Under such conditions, economic grievances generated by automation may not be readily translated into support for redistributive parties.

In particular, the weakening of labor unions disrupts this process of translation, which has historically operated through the organizational mediation of class interests into collective political mobilization (Kitschelt, 2012). Automation in

manufacturing disrupts established patterns of workplace organization, making it more difficult for unions to maintain their central role, while the shift of employment from manufacturing to the non-manufacturing sector further reduces the share of workers who are unionized or effectively reached by unions (Farber, 1987). As this mediating institution weakens, the organizational basis for translating class interests into progressive partisan support erodes, leaving displaced workers without the collective structures that previously anchored their political identity to the left.

The Korean case illustrates how this mechanism is further intensified by specific patterns of labor market and union organization. While overall union density remains relatively low in comparative perspective, Korean unions are highly segmented and concentrated among regular workers in large firms (Peng, 2012; Rueda et al., 2015; Kim, 2017). This segmentation is reinforced by the fact that Korean unions primarily protect the interests of regular workers, while offering limited organizational capacity to incorporate non-regular or more vulnerable workers. Consequently, the costs of automation-related restructuring tend to fall disproportionately on non-regular workers, whereas productivity gains are more likely to accrue to regular employees (Kim & Kim, 2024).

This lack of organizational representation implies that automation-related economic risks are less likely to be collectively articulated in political terms and are instead more likely to be channeled into more conservative social norms and culturally inflected attitudes. Several studies suggest that economic shocks

associated with automation and trade shocks are often accompanied by shifts in political attitudes toward more exclusionary and authoritarian positions (Autor et al., 2020; Bonomi et al., 2021; Wu, 2022; Ferrara, 2023). Experimental findings further show that respondents are more willing to endorse layoffs of immigrant workers when job losses are attributed to automation rather than to generic firm restructuring (Gamez-Djokic & Waytz, 2020). These patterns indicate that economic shocks are not processed purely in material terms but are frequently filtered through culturally salient considerations. This is particularly the case for automation-related shocks, which, even when originating in economic restructuring, are often processed through culturally inflected perceptions that mediate how economic risk translates into political attitudes.

Chapter 3. Data and Methodology

1. Data

We draw on data from several sources to construct our analytical dataset, which can be grouped into three main categories. These categories are electoral data from the National Election Commission of Korea, data used to construct our measure of robot exposure, drawn from the IFR, Census on Establishment (COE), Singapore Department of Statistics, and United Nations Industrial Development Organization (UNIDO), and the regional demographic data from Korean Statistical Information Service (KOSIS).

Our primary outcome variables are derived from electoral data provided by the National Election Commission of Korea, which reports vote counts, turnout figures, and registered voter populations for all elections held in South Korea. The Korean electoral system comprises three major types of elections: presidential elections, National Assembly elections, and local elections. Importantly, presidential election results are reported at the county-level, whereas National Assembly and local elections are reported at the constituency level, which often aggregates multiple cities or divides larger cities into separate

districts. For this reason, we focus exclusively on presidential elections, which provide consistent geographic units that align with our robot exposure measures and allow us to track political outcomes at a granular level over time.

From the presidential election data, we observe vote counts for each party and candidate, the total number of valid votes cast, the number of abstentions, and the total number of registered voters in each region. Following the approach in Autor et al. (2020), we construct two outcome variables. First, we define the change in turnout share as the percentage point change in the ratio of total votes cast to registered voters between consecutive election periods. Second, we define the change in conservative two-party vote share as the percentage point change in the share of the two-party vote won by the conservative party, where the two-party vote share is calculated as the number of votes for the conservative party divided by the sum of votes for the conservative and progressive parties. In this paper, we classify the People Power Party as the conservative party and the Democratic Party as the progressive party. Since presidential elections in Korea are held every five years, our analysis is structured around three five-year periods, 2007-2012, 2012-2017, and 2017-2022, with si-gun-gu as the unit of analysis.

To construct our measure of robot exposure, we use data on the stock of industrial robots by country-industry pairs from the IFR. The IFR compiles data collected from nearly all industrial robot suppliers worldwide, supplemented with information provided by national robot associations. A robot is defined by the IFR

as “an automatically controlled, reprogrammable, multipurpose manipulator, programmable in three or more axes, which can be either fixed in place or mobile, for use in industrial automation applications.” This definition ensures that only true industrial robots, rather than simpler automated machinery, are included in the counts. Industry classifications follow the 19 industry categories based on the framework used in Acemoglu and Restrepo (2020). For robots that are not assigned to a specific industry in the raw data, we allocate them proportionally across industries according to the distribution observed in the classified data. From these data, we calculate the national-level change in robot density, defined as robots per thousand workers, for each industry and period.

We complement these data with employment data by industry and region from the COE. Conducted annually since 1994, the COE is an administrative census of all business establishments in Korea with at least one paid worker. Because the COE measures employment at the establishment level, that is, at each physical place of business rather than at the level of the firm, employment is attributed to the county in which economic activity actually takes place. The COE reports employment counts at the most detailed level of the Korean Standard Industrial Classification, which we aggregate into the 19 IFR industry categories described above. We describe in Chapter 3.2. how these data serve as a key input in constructing both our measure of robot exposure and our instrumental variable.

To construct our instrumental variable, we use employment data from Singapore. For manufacturing industries, we obtain employment data from the

UNIDO Industrial Statistics Database, which classifies employment by industry following the International Standard Industrial Classification (ISIC). We rely on UNIDO for manufacturing industries because its ISIC-based classification corresponds directly to the industry categories used in constructing our IFR-based robot exposure measure, following Klump et al. (2021) and Acemoglu & Restrepo (2020). Because UNIDO does not cover non-manufacturing industries, we obtain employment data for these industries from the Department of Statistics in Singapore, which we map to the corresponding categories. As with the COE data, we aggregate these employment figures into the 19 IFR industry categories described above. Combined with the IFR robot stock data for Singapore, these employment data provide the shift component of our shift-share instrument.

Finally, our control variables for demographic characteristics are obtained from KOSIS, including the share of the population with college education, the share of the population aged 65 or above, the share of manufacturing workers, the share of immigrants, and the net migration rate (defined as the difference between in-migrants and out-migrants divided by total population).

This leaves 227 cities, excluding Jeju and Seogwipo, observed over three five-year periods, 2007–2012, 2012–2017, and 2017–2022, yielding 681 county-period observations. Table 1 presents descriptive statistics for the key variables used in our analysis. The table reports summary statistics for the first-differenced outcome and explanatory variables, as well as for the control variables, which enter our specifications in levels. The mean change in turnout is

4.4 percentage points, with a standard deviation of 5.5 percentage points. The mean change in the conservative two-party vote share is −4.0 percentage points, with a standard deviation of 12.5 percentage points. The mean change in robot exposure is 6.746 robots per thousand workers, with a standard deviation of 9.888, indicating substantial variation across cities in exposure to automation. Our instrument has a mean of 3.458 and a standard deviation of 7.584, in the same units. Turning to the control variables, the average region-period observation has 16.9 percent of its population aged 65 or above and 11.4 percent of its population with a college education.

[Table 1]

2. Empirical Strategy

1) Model

To examine the political effects of automation, we consider the following model:

$$\Delta y_{it} = \beta\, \Delta robot_{it} + X'_{i,t-1}\gamma + \delta_i + \eta_t + \varepsilon_{it} \quad (1)$$

where Δy_{it} is the change in the outcome variable (conservative two-party vote share or turnout) in region i during period t, measured as the difference between

consecutive presidential elections (e.g., $\Delta y_{i,1} = y_{i,2012} - y_{i,2007}$ for period 2007–2012). The explanatory variable, $\Delta robot_{it}$, is the change in robot exposure measured as robots per 1000 workers. We control for baseline demographic characteristics $X_{i,t-1}$ measured at the start of each period, including the share of college graduates and population aged 65 and above. Region-fixed effects δ_i absorb time-invariant local characteristics, such as geographic and institutional factors. Although first-differencing already removes such time-invariant regional levels, including δ_i explicitly accounts for region-specific average changes across periods, which is equivalent to allowing for region-specific linear trends in the underlying level model. By contrast, year-fixed effects η_t are not removed by first-differencing. Common shocks that vary by period, such as nationwide electoral trends or aggregate economic conditions affecting all regions in a given election cycle, remain embedded in Δy_{it} unless explicitly controlled for. We therefore include η_t to absorb these period-specific common changes, and ε_{it} is an idiosyncratic error term.

The coefficient β captures the effect of a one-unit increase in robot exposure, that is, one additional robot per thousand workers over the period, on the percentage point change in the outcome variable between consecutive presidential elections. Whether β in equation (1) admits a causal interpretation depends critically on the exogeneity of regional variation in robot exposure with respect to political outcomes, conditional on the baseline controls, region-fixed effects, and year-fixed effects included in the specification. Even after

accounting for time-invariant regional characteristics and common period-specific trends, the change in robot exposure $\Delta robot_{it}$ may be endogenous if robot adoption in certain industries responds to region-specific political or economic conditions that independently affect voting behavior. We estimate equation (1) using two-stage least squares (2SLS), instrumenting $\Delta robot_{it}$ with a shift-share instrument based on robot adoption in Singapore. We proceed as follows: we first outline how the robot exposure measure is constructed using a shift-share approach, before turning to the identification strategy we employ to isolate the causal effect of automation on political outcomes.

2) Identification Strategy

We construct robot exposure using a shift-share approach following Acemoglu and Restrepo (2020) and Kim (2024). Figure 2 illustrates the geographic distribution of cumulative robot exposure across Korean cities over the full sample period, 2007 to 2022, measured in robots per thousand workers. The map reveals substantial variation across cities, with the highest levels of exposure concentrated in southern Gyeonggi Province, the Chungcheong region, Gyeongsang Province, and parts of Jeolla Province. This spatial pattern reflects the geographic concentration of manufacturing in Korea, driven in large part by the expansion of battery production for hybrid and electric vehicles and the growth of semiconductor and display manufacturing (Kim, 2024).

[Figure 2]

We now describe formally how this measure is constructed. Specifically, for region i and period t, robot exposure is defined as:

$$\Delta robot_{it} = \sum_{j=1}^{J} \frac{emp_{ij,2005}}{emp_{i,2005}} \times \frac{\Delta robot_{j,t}}{emp_{j,2005}} \quad (2)$$

where $\frac{emp_{ij,2005}}{emp_{i,2005}}$ is the employment share of industry j in region i in 2005, $\Delta robot_{j,t}$ is the change in the stock of robots in industry j over period t in Korea, and $emp_{j,2005}$ is national employment in industry j in 2005. $\Delta robot_{it}$ captures the predicted increase in robots per 1000 workers in region i based on its initial industrial composition and national trends in robot adoption by industry. The subscript t indexes the three five-year periods corresponding to consecutive presidential elections, 2007–2012, 2012–2017, and 2017–2022, and $\Delta robot_{j,t}$ is computed as the first difference in the robot stock over each period, which are then stacked to form our panel.

Although we control for baseline regional characteristics and include region and year fixed effects, the robot exposure measure may remain correlated with time-varying regional confounders that independently affect political outcomes. Both components of the shift-share measure, the regional employment shares and the industry-level shift, are potential sources of endogeneity.

The 2005 employment shares may be endogenous if firms' location decisions

already reflected anticipation of future automation or political trends. Korea's robotics expansion began around 2005–2006 (Adema et al., 2025), and firms in automation-intensive industries may have strategically concentrated in regions favorable to robot adoption. To address this concern, our instrument instead uses 1995 employment shares (equation (2)), ensuring a 12-year lag that makes anticipation implausible.

Korean industry-level robot adoption may be endogenous if domestic industry-specific conditions simultaneously drive both robot adoption and regional political outcomes. For example, government industrial policies promoting specific sectors may increase robot adoption while directly affecting regional employment and voting behavior. Similarly, industry-specific labor pressures or economic shocks may independently influence both automation decisions and political preferences. In these cases, we cannot distinguish whether political changes result from robot-induced disruptions or from the underlying domestic conditions, such as industrial policies, labor market pressures, or economic shocks that drove robot adoption in the first place.

To address this concern, we instrument Korean robot adoption with Singaporean robot adoption, which reflects global technological trends but is orthogonal to Korea-specific domestic conditions (Kim, 2024). The validity of this instrument rests on the similarity in industrial composition between the two economies. As shown in Figure 3, Korea and Singapore share a comparable distribution of employment across industries. The correlation coefficient between

industry-level employment shares in the two countries is 0.970 (p-value < 0.001), confirming that global technological forces driving robot adoption in Singapore translate into similar adoption pressures in Korea, independently of any Korea-specific factors.

[Figure 3]

To further address the potential endogeneity of regional employment shares used as weights, we follow Acemoglu and Restrepo (2020) and use pre-determined 1995 employment shares in place of contemporaneous shares. Our instrument is thus constructed as:

$$\Delta robot_{it}^{IV} = \sum_{j=1}^{J} \frac{emp_{ij,1995}}{emp_{i,1995}} \times \frac{\Delta robot_{j,t}^{SG}}{emp_{j,2005}^{SG}} \quad (3)$$

where $\frac{emp_{ij,1995}}{emp_{i,1995}}$ is the 1995 composition of employment across industries and regions, $\Delta robot_{j,t}^{SG}$ is the change in the stock of robots in industry j in Singapore over period t, and $emp_{j,2005}^{SG}$ is the national employment in industry j in Singapore in 2005, used to normalize the robot stock change into robots per thousand workers.

We use 1995 employment shares for several reasons. First, following Kim (2024), 1995 is the earliest year for which data are available, and robot density in Korea was effectively zero at that time, ensuring that the 1995 industrial composition predates the automation wave entirely. Second, the 10-year gap

between 1995 and 2005 spans multiple structural breaks in Korea's economy and political system. The East Asian Financial Crisis (1997–1998) triggered massive industrial restructuring that fundamentally reshaped Korea's employment landscape. Additionally, Korea's first opposition party victory in the 1997 presidential election marked a critical political realignment, consolidating the democratic transition that began in 1987 and establishing new patterns of regional political competition that would define subsequent elections. The 1995 employment distribution therefore predates both this political watershed and the automation trends of the 2000s–2010s, ensuring our share component captures historical industrial geography rather than forward-looking responses to emerging political or technological trends.

Table 2 presents the first-stage results, demonstrating that our instrument serves as a relevant instrument for the change in robot exposure in Korea. Column (1) shows a parsimonious specification without any control variables, yielding a coefficient of 0.190 (standard error: 0.023). Column (2) adds region-fixed effects and year-fixed effects, which increases the coefficient to 0.156 (standard error: 0.031). Column (3) further incorporates demographic controls, resulting in a coefficient of 0.124 (standard error: 0.034). The first-stage F-statistics are consistently large across all specifications, alleviating concerns about weak instruments.

[Table 2]

Additionally, we further assess the validity of the exclusion restriction by conducting a pre-trend test. Table 3 regresses pre-period political outcomes over 2002-2007 on our instrument constructed using the 2007-2012 change in robot exposure in Singapore. All coefficients in Table 3 are close to zero, ranging from -0.000 to 0.003, and remain statistically insignificant. These null findings indicate that the instrument does not correlate with pre-existing political patterns, supporting the validity of the IV approach.

[Table 3]

Chapter 4. Results

1. Main Results

Panel C of Table 4 presents the FE-IV estimates of the effects of robot exposure on political behavior, distinguishing between change in voter turnout and conservative vote share. First, results from all specifications indicate that the IV is strong. We also report the Anderson-Rubin F-statistics and the corresponding p-values to test whether the estimated effects on political behavior are robust to weak identification (Anderson & Rubin, 1949). The second-stage estimates indicate that robot exposure significantly affects both voter turnout and conservative vote share. The magnitude of the effects indicates that an increase of one robot per thousand workers leads to a statistically significant 1.4 percentage point decline in voter turnout and a 2.3 percentage point increase in conservative vote share.

[Table 4]

These findings are broadly consistent with the pattern documented in prior

work on automation and political behavior (Frey et al., 2018; Anelli et al., 2019; Gonzalez-Rostani, 2024). Gonzalez-Rostani (2024) shows that automation exposure reduces political participation, while Anelli et al. (2019) and Frey et al. (2018) document that regions more exposed to robot adoption shift toward conservative parties in Western Europe and the United States respectively. This consistency is notable given that South Korea's political institutions and labor market structures differ substantially from those of the Western democracies in which these effects were first documented. That directionally similar effects emerge nonetheless suggests that the political consequences of automation may generalize beyond the specific settings in which they have been most extensively studied.

For comparison, Panels A and B of Table 4 present pooled OLS and fixed effects estimates, respectively. The pooled OLS estimates in Panel A show small negative coefficients on both turnout (−0.001) and conservative vote share (−0.001). Adding region and year fixed effects in Panel B does not substantially alter the results: both coefficients remain negative, though the conservative vote share coefficient loses statistical significance. The FE-IV estimates in Panel C are substantially larger in magnitude for both outcomes and reverse the sign for conservative vote share, suggesting that classical measurement error in robot exposure severely attenuates non-IV estimates toward zero.

Table 5 presents our main results with additional demographic controls to address potential confounding from regional population characteristics. We

include the share of population aged 65 and over and the share of college graduates as proxies for local demographic composition and skill structure, which may independently affect both automation adoption and political preferences. The second-stage estimates remain consistent with our baseline findings. Robot exposure significantly reduces voter turnout by 1.4 percentage points and increases conservative vote share by approximately 2.6 percentage points. The Kleibergen-Paap F-statistics exceed 10 in all specifications, confirming instrument strength. Moreover, the Anderson-Rubin F-test, which is robust to weak identification, yields p-values below 0.01 in all cases, further validating our inference. These results indicate that the estimated effects of automation on political behavior are robust to controlling for age structure and educational composition.

[Table 5]

We also examine whether the effects persist when restricting the sample to competitive regions, defined as cities where neither party consistently received more than 55 percent of the two-party vote share across all electoral periods from 2007 to 2022 (Autor et al., 2020). Columns (3) and (4) of Table 5 report the results. The estimated effects remain statistically significant and consistent in direction with the full-sample results. While the magnitudes are somewhat smaller relative to the full sample, robot exposure continues to significantly

reduce voter turnout and increase conservative vote share.

To examine whether the political effects of automation vary across demographic contexts, we conduct a heterogeneity analysis by immigrant share. We classify regions as high or low immigrant share based on whether the share of immigrants in 2007 is above or below the sample median. Table 6 presents the FE-IV estimates separately for these two groups. In high immigrant share regions, robot exposure significantly reduces voter turnout by 1.3 percentage points and increases conservative vote share by 2.2 percentage points. In contrast, effects in low immigrant share regions are statistically insignificant for both outcomes[1]. These findings suggest that the political effects of automation are concentrated in regions with higher immigrant presence.

[Table 6]

These findings are consistent with the possibility that the local presence of immigrant workers intensifies the economic insecurity generated by automation. Workers in these regions may face layered displacement pressures from two directions simultaneously, competition from immigrant labor on one side and substitution by machines on the other, a substitutability documented using

[1] We acknowledge that the Kleibergen-Paap F-statistics are substantially reduced in the subgroup analysis, particularly in low immigrant share regions, reflecting the loss of first-stage precision from sample splitting. We therefore rely on the Anderson-Rubin test for inference, which is robust to weak identification. The Anderson-Rubin F-statistics remain large and statistically significant across all columns ($p < 0.01$), supporting the validity of our estimates.

establishment-level data from South Korea (Ahn, 2026). This layered pressure is likely to generate more acute and persistent economic distress than automation exposure alone.

Economic distress of this kind imposes a psychological burden that depletes cognitive resources. Several strands of literature show that financial scarcity redirects cognitive attention toward immediate material concerns, crowding out cognitive capacity for longer-term deliberation (Mani et al., 2013; Mullainathan & Shafir, 2013). Political participation, which requires gathering information, forming judgments, and turning out to vote, demands precisely this kind of deliberate engagement. Workers in high-immigrant regions may face compounded economic pressure from both automation-induced displacement and heightened labor market competition, and to the extent that this dual pressure further depletes cognitive resources, it offers one plausible mechanism for the amplified voter turnout decline observed in these regions.

With respect to conservative vote share, a further possibility is that the economic insecurity generated in these settings is channeled, at least in part, through attitudes toward immigration. This is because the two sources of pressure are not necessarily perceived symmetrically (Gamez-Djokic & Waytz, 2020; Wu, 2022; Gonzalez-Rostani, 2024). Technological change is a diffuse and less legible process, whereas immigrant workers represent a more visible and identifiable presence in the local labor market. Indeed, survey evidence shows that workers facing higher automation risk are systematically more likely to

oppose immigration and free trade, suggesting that automation-induced economic anxiety tends to be directed toward more tangible and proximate targets rather than toward the underlying technological process itself (Gamez-Djokic & Waytz, 2020; Wu, 2022). In regions where immigrants are more concentrated, this asymmetry is more likely to be activated, channeling economic distress into anti-immigrant sentiment and, in turn, into support for conservative parties.

2. Robustness Checks

While region and year fixed effects absorb time-invariant confounders, time-varying regional characteristics correlated with both robot exposure and political outcomes remain a concern. We therefore include two sets of additional covariates. First, we include the immigrant share, the manufacturing employment share, and the net migration rate, each measured at the beginning of the relevant period (2007, 2012, and 2017). Second, to address concerns about differential pre-existing trends, we include the lagged change in each outcome variable as a pre-trend control in columns (3) and (4) of Table 7. Specifically, for each period, we include the change in the outcome variable from the immediately preceding period.

[Table 7]

Table 7 presents the results. Across all specifications, the estimated effects remain statistically significant and broadly consistent in magnitude with our baseline estimates, suggesting that our main findings are not driven by time-varying demographic composition or differential pre-existing trends[2].

[2] While the Kleibergen-Paap F-statistics fall slightly below the conventional threshold of 10 in some specifications, reflecting reduced first-stage precision from the inclusion of additional covariates, the Anderson-Rubin test remains highly significant across all columns ($p < 0.01$), providing valid inference robust to weak identification.

Chapter 5. Conclusion

This paper investigates how exposure to industrial robots affects political behavior in South Korea, using county-level data from presidential elections between 2007 and 2022. We construct a shift-share instrumental variable that interacts national-level robot adoption trends, proxied by the industry-specific robot adoption patterns in Singapore, with the industrial composition of each city. This design allows us to isolate variation in local robot exposure that is plausibly unrelated to confounding local shocks, while still capturing the heterogeneous impact of a common technological trend across regions with different industrial structures.

Our estimates indicate that increased robot exposure significantly reduces voter turnout and increases the vote share of the conservative party. These average effects, however, mask substantial heterogeneity. The effects on both voter turnout and conservative vote share are statistically significant only in cities with larger immigrant populations, while the effects are insignificant in cities with lower immigrant concentration. This pattern suggests that the political effects of automation are amplified where labor market displacement risk is most acute and where immigrant labor may intensify economic insecurity among native

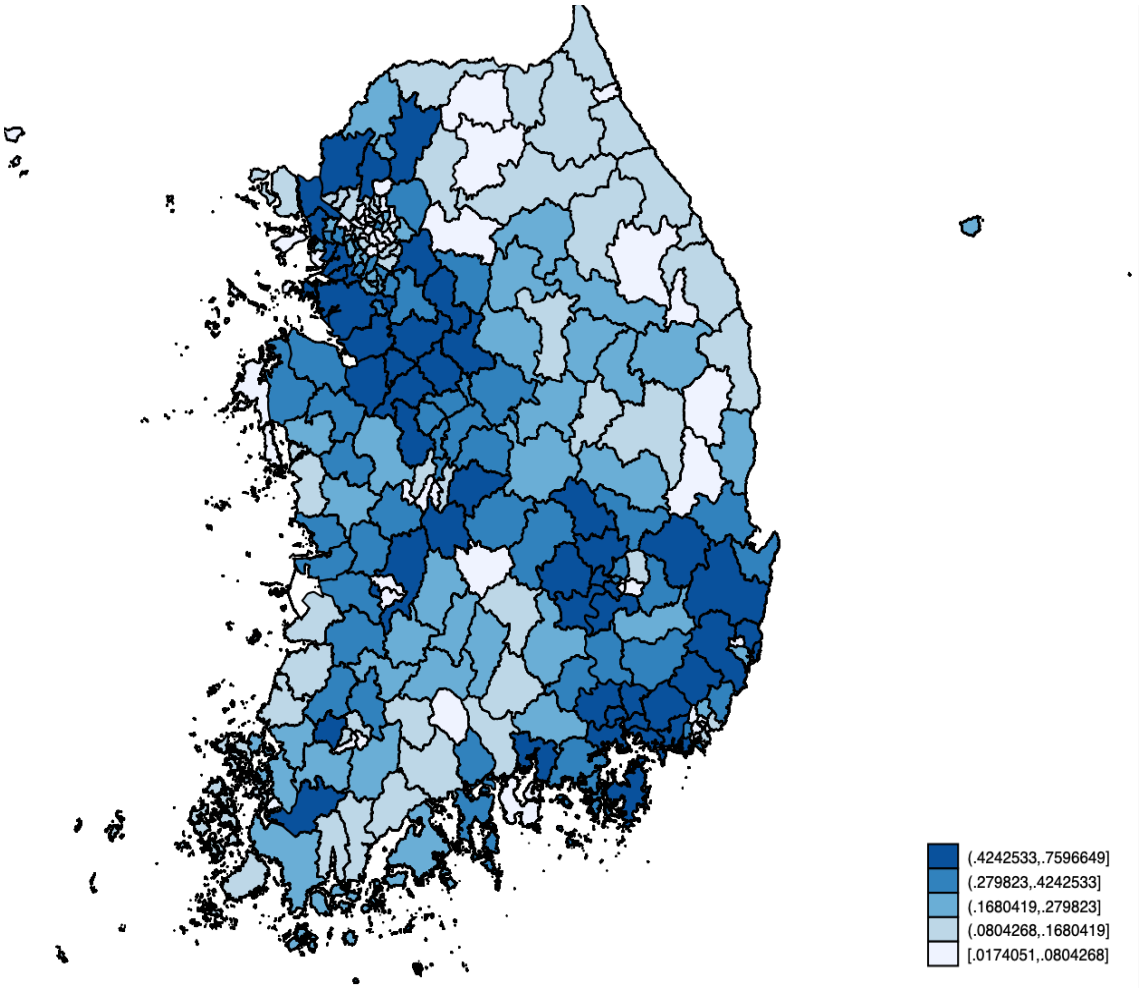
workers, conditions that may in turn heighten psychological and cultural responses to economic threat.

These findings contribute to a growing literature on the political consequences of economic disruption. Prior work has documented that trade shocks, globalization, and immigration shape political behavior in important ways (Swank & Betz, 2003; Autor et al., 2020; Tabellini, 2020), yet the political consequences of technological change have received comparatively less attention despite its arguably greater role in labor market disruption. Our results provide the first causal evidence that automation, measured through plausibly exogenous variation in robot exposure, significantly shapes both electoral participation and partisan preferences in Korea. Moreover, the fact that a similar pattern emerges in South Korea suggests that the political consequences of technological change may reflect a more general response to localized economic insecurity, rather than features specific to Western institutional contexts. As automation technologies continue to advance and as artificial intelligence threatens to extend displacement pressures beyond the workers in the specific industries most affected by industrial robotics, understanding how technological change shapes political behavior across diverse institutional settings becomes an increasingly pressing question.

Several limitations warrant caution in interpreting these results. Our analysis relies on county-level aggregate data, which precludes a direct test of the individual-level mechanisms that plausibly underlie the observed patterns.

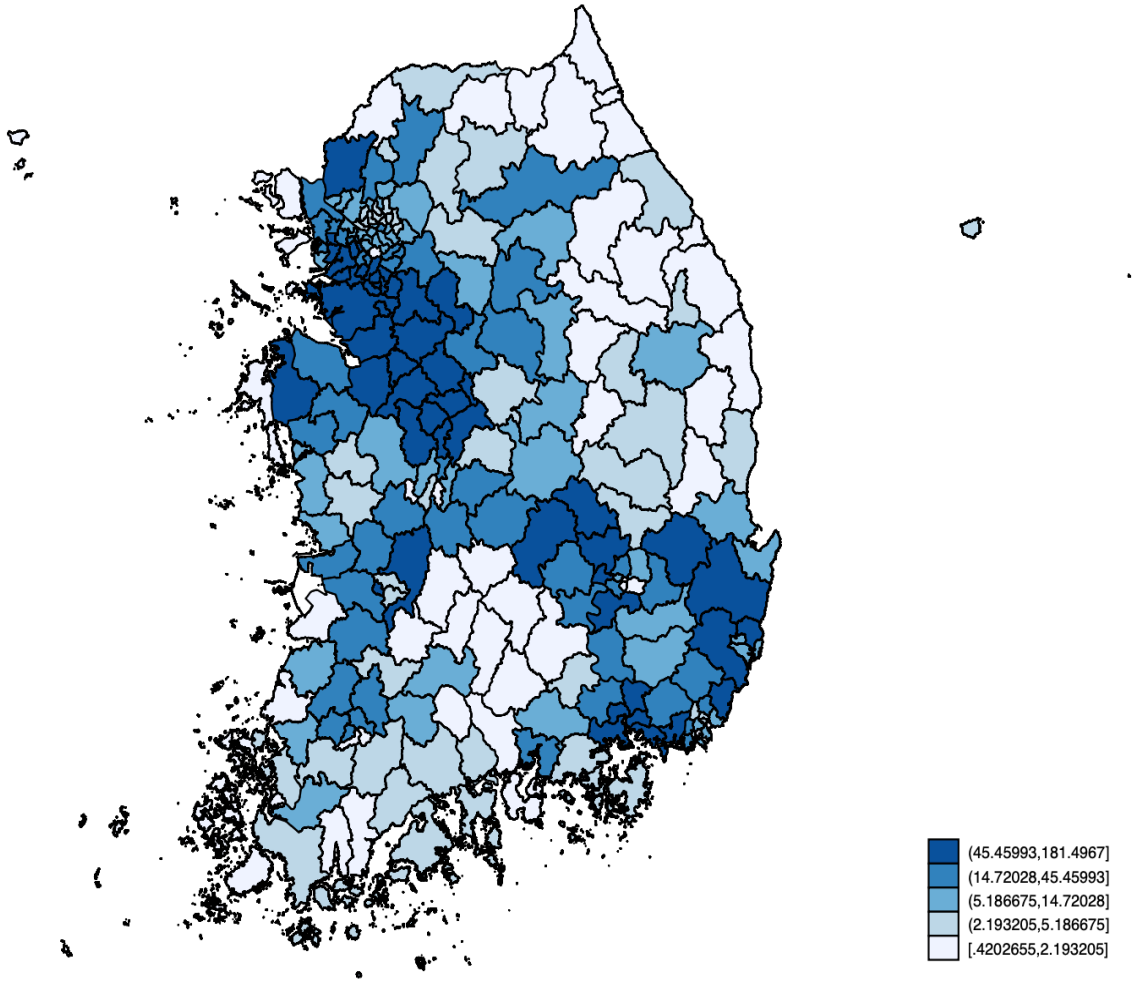
Whether the effects operate through resource depletion, psychological responses to economic threat, or shifts in cultural attitudes toward immigration remains an open question that aggregate data alone cannot resolve. Individual-level panel data would allow for more direct testing of these mechanisms, and comparative analysis across additional non-Western democracies would help establish whether the patterns documented here generalize beyond the Korean case.

Figure 1. Geographic Distribution of Manufacturing Employment Share, 2007



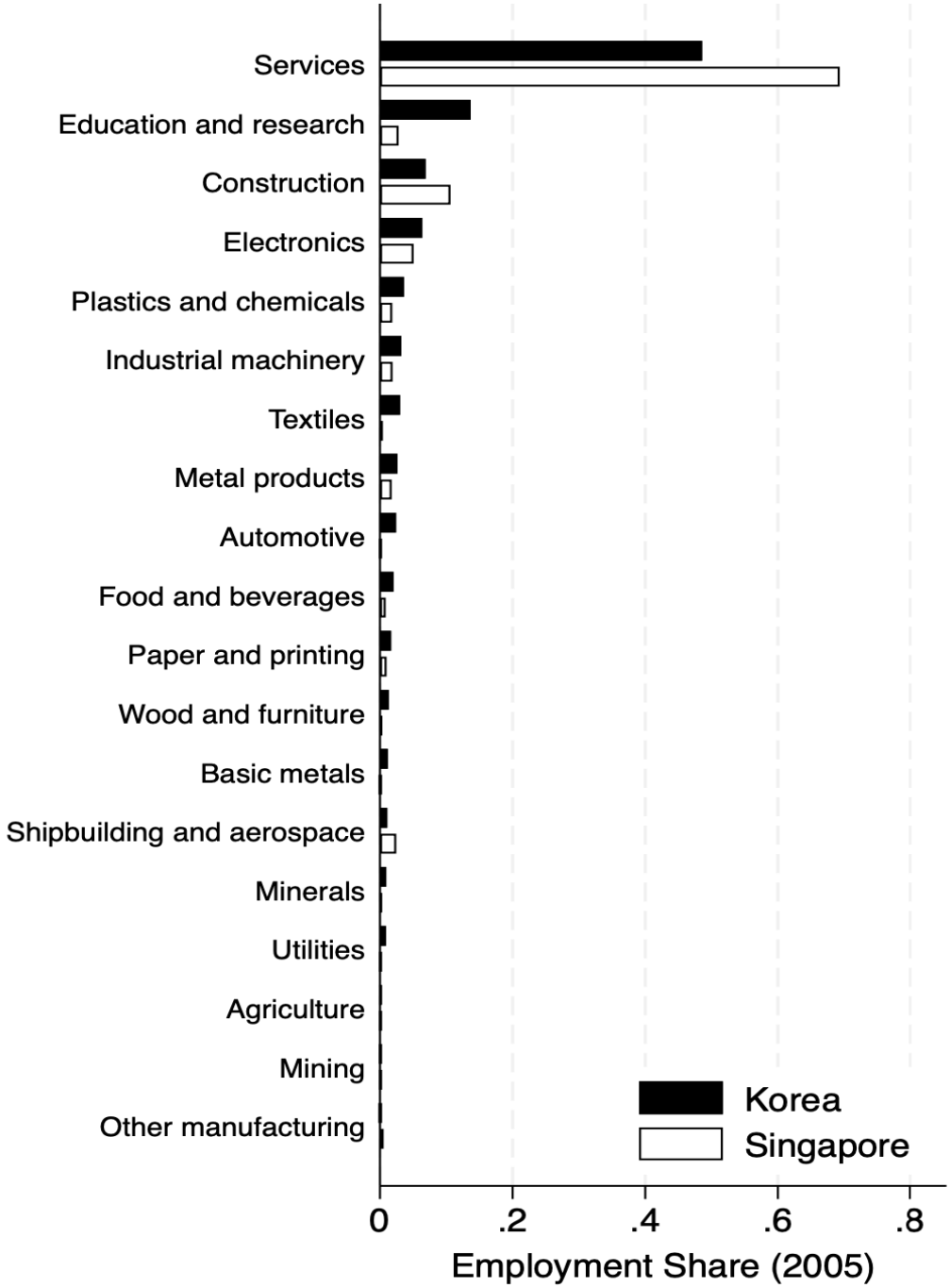
Notes: Data are drawn from the COE. Manufacturing share is defined as the share of manufacturing workers in total employment in 2007, where industries are classified following Acemoglu and Restrepo (2020).

Figure 2. Geographic Distribution of Change in Robot Exposure, 2007–2022



Notes: The map displays the long difference in robot exposure from 2007 to 2022, measured in robots per thousand workers.

Figure 3. Industry-Level Employment Share: Korea and Singapore



Notes: Industry employment shares, defined as the share of workers employed in each industry relative to total employment, are computed using COE data for Korea and data drawn from UNIDO and Singapore Department of Statistics.

Table 1. Summary Statistics

Variables	Observations	Mean	Std. Dev
Turnout share	681	0.044	0.055
Conservative two-party vote share	681	-0.040	0.125
Robot exposure	681	6.746	9.888
Shift-share instrument	681	3.458	7.584
Share of population aged 65+	681	0.169	0.078
Share of college-educated	681	0.114	0.063

Notes: Turnout, conservative two-party vote share, robot exposure, and the instrument are first-differenced variables, defined as the change between consecutive presidential election periods. The two control variables, population aged 65 or above and college-educated population, are level variables measured as shares at the start of each period.

Table 2. First-stage Relationship, 2007-2022

Dependent variable: Change in robot exposure, 2007-2022	(1)	(2)	(3)
Robot exposure from Singapore	0.190***	0.156***	0.124***
(2007-2022)	(0.023)	(0.031)	(0.034)
F-statistic	67.445	175.801	102.473
Region FEs	Yes	Yes	Yes
Year FEs		Yes	Yes
Control variables			Yes
Observations	681	681	681

Notes: *** p<0.01, ** p<0.05, * p<0.10. Standard errors are clustered at the county level. Control variables include the share of population aged 65 or above and the share of college-educated population, measured in levels at the start of each period.

Table 3. Pre-trend Check

	(1) Turnout	(2) Conservative	(3) Turnout	(4) Conservative
Robot exposure from Singapore (2007-2012)	-0.000	0.003	0.000	0.001
	(0.001)	(0.002)	(0.001)	(0.002)
Control variables			Yes	Yes
Observations	227	227	227	227

Notes: *** p<0.01, ** p<0.05, * p<0.10. The dependent variables are the changes in turnout and conservative two-party vote share over the pre-sample period 2002–2007. The independent variable is the shift-share instrument constructed using robot adoption in Singapore over 2007–2012. Standard errors are clustered at the county level. Control variables include the share of population aged 65 or above and the share of college-educated population, measured in levels at the start of each period.

Table 4. The Effects of Robot Exposure on Voter Turnout and Conservative Vote Share

	(1) Turnout	(2) Conservative
Panel A. Pooled OLS		
Robot exposure	-0.001***	-0.001***
	(0.000)	(0.000)
Panel B. FE estimation		
Robot exposure	-0.001**	-0.001
	(0.000)	(0.000)
Panel C. FE-IV estimation		
Second stage		
Robot exposure	-0.014***	0.023***
	(0.003)	(0.007)
First stage		
IV	0.156***	0.156***
	(0.031)	(0.031)
C-D F-stat	18.204	18.204
K-P F-stat	25.628	25.628
A-R F-stat	72.123	28.816
A-R p-value	0.000	0.000
Observations	681	681

Notes: C-D F-stat: Cragg-Donald F-statistic. K-P F-stat: Kleibergen-Paap F-statistic. A-R F-stat: Anderson-Rubin F-statistic. Standard errors are clustered at the county level and reported in parentheses. Panel A reports pooled OLS estimates without fixed effects or control variables. Panel B adds region and year fixed effects. Panel C reports FE-IV estimates with region and year fixed effect. *** p<0.01, ** p<0.05, * p<0.10.

Table 5. Robot Exposure and Political Outcomes: Baseline Extensions

	Full sample		Only competitive region	
	(1) Turnout	(2) Conservative	(3) Turnout	(4) Conservative
Second stage				
Robot exposure	-0.014***	0.026***	-0.008***	0.013**
	(0.005)	(0.010)	(0.003)	(0.006)
First stage				
IV	0.124***	0.124***	0.143***	0.143***
	(0.034)	(0.034)	(0.032)	(0.032)
C-D F-stat	9.675	9.675	7.510	7.510
K-P F-stat	12.943	12.943	19.966	19.966
A-R F-stat	48.395	23.008	19.837	7.072
A-R p-value	0.000	0.000	0.000	0.009
Region FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Control variables	Yes	Yes	Yes	Yes
Observations	681	681	405	405

Notes: C-D F-stat: Cragg-Donald F-statistic. K-P F-stat: Kleibergen-Paap F-statistic. A-R F-stat: Anderson-Rubin F-statistic. Standard errors are clustered at the county level and reported in parentheses. Control variables include the share of population aged 65 or above and the share of college-educated population, measured in levels at the start of each period. *** p<0.01, ** p<0.05, * p<0.10.

Table 6. Heterogeneity Analysis by Immigrant Share

	High immigrant share		Low immigrant share	
	(1) Turnout	(2) Conservative	(3) Turnout	(4) Conservative
Robot exposure	-0.013**	0.022*	-0.046	0.087
	(0.005)	(0.011)	(0.036)	(0.062)
C-D F-stat	6.880	6.880	0.403	0.403
K-P F-stat	8.162	8.162	1.514	1.514
A-R F-stat	29.380	9.959	11.301	14.526
A-R p-value	0.000	0.002	0.001	0.000
Region FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Control variables	Yes	Yes	Yes	Yes
Observations	342	342	339	339

Notes: C-D F-stat: Cragg-Donald F-statistic. K-P F-stat: Kleibergen-Paap F-statistic. A-R F-stat: Anderson-Rubin F-statistic. Regions are classified as high (low) immigrant share if the share of immigrants in 2007 is above (below) the sample median. While Kleibergen-Paap F-statistics are reduced due to sample splitting, the Anderson-Rubin test remains significant across all columns, providing valid inference robust to weak identification. Control variables include the share of population aged 65 or above and the share of college-educated population, measured in levels at the start of each period. Standard errors are clustered at the county level. *** p<0.01, ** p<0.05, * p<0.10.

Table 7. Robustness Checks

	(1) Turnout	(2) Conservative	(3) Turnout	(4) Conservative
Robot exposure	-0.016**	0.022**	-0.016***	0.019*
	(0.006)	(0.011)	(0.006)	(0.011)
C-D F-stat	7.781	7.781	9.318	6.820
K-P F-stat	8.224	8.224	9.470	7.333
A-R F-stat	38.381	13.596	46.363	11.782
A-R p-value	0.000	0.000	0.000	0.001
Region FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes	Yes
Immigrant share	Yes	Yes	Yes	Yes
Manufacturing share	Yes	Yes	Yes	Yes
Net migration rate	Yes	Yes	Yes	Yes
Pre-trend controls			Yes	Yes
Observations	681	681	681	681

Notes: C-D F-stat: Cragg-Donald F-statistic. K-P F-stat: Kleibergen-Paap F-statistic. A-R F-stat: Anderson-Rubin F-statistic. All specifications include region and year fixed effects. Baseline controls include the share of population aged 65 or above and the share of college-educated population. Columns (1) and (2) additionally include immigrant share, manufacturing share, and net migration rate, measured in levels at the start of each period. Columns (3) and (4) further add pre-trend controls, defined as the change in each outcome variable during the immediately preceding period. While Kleibergen-Paap F-statistics fall slightly below the conventional threshold in some specifications, the Anderson-Rubin test remains significant across all columns ($p < 0.01$), providing valid inference robust to weak identification. Standard errors are clustered at the county level. *** p<0.01, ** p<0.05, * p<0.10.

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논문요약

로봇 도입과 투표 행태

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본 연구는 자동화가 정치 참여 및 정당 선호에 끼치는 영향을 분석한다. 종속변수는 2007년부터 2022년까지 실시된 한국 대통령 선거의 시군구 수준 투표율 및 보수정당 득표율로 측정하고, 로봇 노출도는 각 지역의 산업 구성에 기반한 변이할당 도구변수 접근법을 적용하여 측정되었다. 분석 결과, 로봇 노출도 증가는 투표 참여를 유의미하게 감소시키고 보수 정당에 대한 지지를 증가시키는 것으로 나타나며, 이같은 효과는 이민자 비율이 높은 지역에서 집중적으로 나타난다. 본 연구의 결과는 기술변화가 정치에 미치는 영향에 관한 한국 맥락에서 최초의 인과적 증거를 제시하며, 자동화가 서구 맥락에서 확인된 문화적 및 심리적 메커니즘과 유사한 방식으로 한국의 선거 참여 및 정당 선호가 형성될 수 있음을 시사한다.

주제어: 자동화, 기술변화, 투표 행태, 정치경제학, 노동시장